

Image Processing



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You are not allowed to use any ready-made image-processing functions. You must perform all operations manually using NumPy and ... and your own mathematical implementations.

Question 1: Object Boundary Extraction Using Morphological Operations

Problem Statement

Use your own binary image containing several objects, extract the boundary of each object using morphological erosion.

Suggested Method

Use the following relation:

$$\text{Boundary} = A - (A \ominus B)$$

Where:

- A : the original binary image
- B : the structuring element
- $\ominus$: the erosion operation

Expected Outputs

- The image containing the extracted boundaries of the objects
- The number of boundary components in the image

Educational Notes

- Understand the difference between **internal and external boundaries**.
- Analyze how the **size of the structuring element** affects the **thickness of the boundary**.

Question 2: Image Segmentation Using Thresholding and Region Merging

Problem Statement

Implement an image segmentation system that separates an image into distinct regions using **thresholding** and **region merging techniques**.

You must implement the required steps **manually** without using built-in functions that perform these operations automatically (such as `skimage.segmentation`).

Requirements

- The implementation must be done in **Python or MATLAB**.
- Only **basic image processing and array operations** are allowed.

The following functions are **not allowed**:

- Automatic thresholding functions such as:
 - `cv2.threshold`
 - `imbinarize`
- Automatic connected-component labeling functions such as:
 - `bwlabel`
 - `scipy.ndimage.label`

Input

- A **grayscale or RGB image**

Output

- A **labeled image** in which each segmented region is assigned a **unique integer label**.

Objectives

1. Compute the **optimal threshold using Otsu's method** manually.
2. Apply the threshold to create a **binary image**.
3. Implement **connected component labeling manually**.
4. Perform **region merging** for neighboring regions whose **mean intensity difference is below a specified threshold**.

Allowed Libraries

Python:

- `numpy`
- `matplotlib.pyplot`
- `cv2.imread` (for reading images)
- `cv2.cvtColor` (for converting to grayscale)

MATLAB:

- Standard functions for **reading and displaying images**
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Question 3: Cell Counting and Feature Extraction in Grayscale Images

Problem Statement

Design and implement a function to **count cells in grayscale microscopy images** (similar to the provided sample image) **without using automatic labeling functions such as `bwlabel`**.

Part (a)

Implement a function that **detects and counts cells** in a grayscale image.

Part (b)

Write a function whose inputs are:

- the **cell image**
- the **name and path of an Excel file**

The function must generate an **Excel file** containing information for each detected cell.

For every cell, the following information must be stored in a separate row:

- **Cell area**
- **Mean intensity value of the cell**

Example Excel Output Structure

Cell ID Area Mean Intensity

1	...	...
2	...	...
3	...	...

Restrictions

- Do not use automatic connected-component labeling functions such as `bwlabel` or equivalent tools.

Deliverables:

For each question, your submission must include:

- The relevant **mathematical formulation**
- A clear **conceptual explanation**
- The **Python implementation using NumPy and simple libraries.**
- The **resulting output or images**

END
